

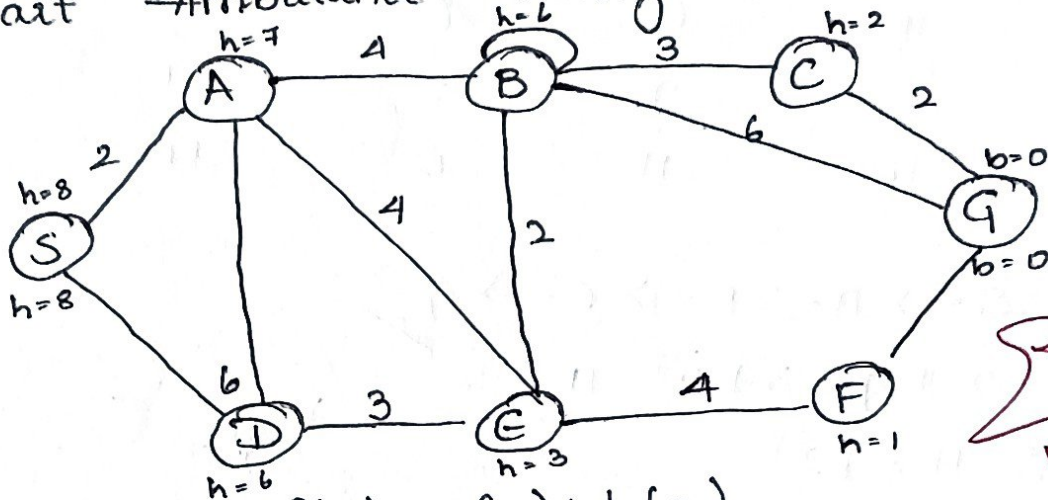
MLAD101 - Artificial Intelligence & Expert Systems

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COR AT2 - Problem Solving Assignment.

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1. Smart Ambulance Navigation - A* Search



AG

10/18/26

From S

$S \rightarrow A = 2$

$$= 2 + 7 = 9 \checkmark$$

$$g(B) = 2 + 4 = 6$$

$$f(B) = 6 + 6 = 12$$

$$g(D) = 2 + 6 = 8$$

$$f(D) = 8 + 6 = 14$$

$$g(E) = 2 + 4 = 6$$

$$f(E) = 6 + 3 = 9 \checkmark$$

$$g(F) = 6 + 4 = 10$$

$$f(F) = 10 + 1 = 11 \checkmark$$

$F \rightarrow G$

$$g(G) = 10 + 2 = 12$$

$$h(G) = 0$$

$$f(G) = 12$$

$S \rightarrow A \rightarrow E \rightarrow F \rightarrow G$

Step	Node	$g(n)$	$h(n)$	$f(n) = g(n) + h(n)$
1	S	0	8	8
2	A	2	7	9
3	E	6	3	9
4	F	10	1	11
5	B	6	6	12
6	C	9	2	11
7	G	11	0	11

$S \rightarrow A \rightarrow B \rightarrow C \rightarrow G$

$$2 + 2 + 3 + 2 = 11$$

$\therefore 11 < 12$

So, the route selected using the given heuristic is not optimal

Reason

The given heuristic is not fully admissible
 for ex:- S the actual cheapest cost to G is 7
 but: $h(S) = 8 > 7$

Explored sequence: $S \rightarrow A \rightarrow E \rightarrow F \rightarrow G$

Selected path cost = 12

Actual optimal path: $S \rightarrow A \rightarrow B \rightarrow C \rightarrow G$

Optimal cost = 11

2. Autonomous warehouse Robot - Online Search.

Given grid

$$S = (5, 1), \quad G = (1, 7)$$

5	S	?	?	?	?	?	?
4	?	#	?	?	#	?	?
3	?	?	?	#	?	?	#
2	?	#	?	?	?	#	?
1	?	?	#	?	?	?	G
	1	2	3	4	5	6	7

Strategy: Online - Greedy Best first search using Manhattan distance

$$h(n) = |x_n - x_g| + |y_n - y_g|$$

Goal: $G = (7, 1)$

Step

Robot position

Decision / Observation

1. $S(1, 5)$
2. $(2, 5)$
3. $(3, 5)$
4. $(4, 5)$
5. $(5, 5)$
6. $(6, 5)$
7. $(7, 5)$
8. $(7, 4)$
9. $(7, 4)$

move right $\rightarrow (2, 5)$

Move right $\rightarrow (3, 5)$

Move right $\rightarrow (4, 5)$

Move right $\rightarrow (5, 5)$

Move right $\rightarrow (6, 5)$

Move right $\rightarrow (7, 5)$

Move down $\rightarrow (7, 4)$

Discovers obstacle at $(7, 3)$

cannot move down

Replan: move left $\rightarrow (6, 4)$

10 (6, 4)

11 (6, 3)

12 (6, 3)

13 (5, 3)

14 (5, 2)

15 (6, 1)

16 (6, 1)

Move down $\rightarrow$ (6, 3)

Discover obstacle at (6, 2)

Cannot move down

Replan: move left $\rightarrow$ (5, 3)

Move down $\rightarrow$ (5, 2)

Move down $\rightarrow$ (5, 1)

Move right $\rightarrow$ (6, 1)

Move right $\rightarrow$ G (7, 1)

final path: S $\rightarrow$ (2, 5) $\rightarrow$ (3, 5) $\rightarrow$ (4, 5) $\rightarrow$
(5, 5) $\rightarrow$ (6, 5) $\rightarrow$ (7, 5) $\rightarrow$ (7, 4) $\rightarrow$ (6, 4)
 $\rightarrow$ (6, 3) $\rightarrow$ (5, 3) $\rightarrow$ (5, 2)

Total moves = 14

When an obstacle is discovered
Detect obstacle $\rightarrow$ Update map $\rightarrow$ Recalculate
 $h(n) \rightarrow$ choose new move

Online search vs Offline search

Online search makes decisions using currently known information & replans when new obstacles appear. Offline search assumes the complete map is known beforehand.

3. University Examination Scheduling - CSP

Given six examinations

$E_1, E_2, E_3, E_4, E_5, E_6$

four time slots: T_1, T_2, T_3, T_4 .

(a) Variables:

$E_1, E_2, E_3, E_4, E_5, E_6$

Domain of each variable

$$D(E_i) = \{T_1, T_2, T_3, T_4\}$$

Student Conflict constraint

$$E_1 \neq E_2, E_1 \neq E_3, E_2 \neq E_4, E_2 \neq E_5, E_3 \neq E_5, \\ E_4 \neq E_6$$

Hall Constraints

Exam	Students	Possible halls
E_1	100	A
E_2	80	A, B
E_3	$\neq 0$	A, B, C
E_4	60	A, B, C
E_5	90	A, B
E_6	50	A, B, C

(b) MRV & forward checking

$$E_2 = T_2$$

Variable

Variable	Remaining domain
E_1	T_1, T_3, T_4
E_2	T_2
E_3	T_1, T_2, T_3, T_4
E_4	T_1, T_3, T_4
E_5	T_1, T_3, T_4
E_6	T_1, T_2, T_3, T_4

Step 2: $E_1 = T_1$

Variable

E_1

E_2

E_3

E_4

E_5

E_6

Remaining domain

T_1

T_2

T_2, T_3, T_4

T_1, T_3, T_4

T_1, T_3, T_4

T_1, T_2, T_3, T_4

Step 3: $E_3 = T_2$

Variable

E_1

E_2

E_3

E_4

E_5

E_6

Remaining domain

T_1

T_2

T_2

T_1, T_3, T_4

T_1, T_3, T_4

T_1, T_2, T_3, T_4

Step 4: $E_4 = T_1$

Variable

E_1

E_2

E_3

E_4

E_5

E_6

Remaining domain

T_1

T_2

T_2

T_1

T_1, T_3, T_4

T_2, T_3, T_4

Step 5: $E_5 = T_3$

Variable

E_1

E_2

E_3

E_4

E_5

E_6

Remaining domain

T_1

T_2

T_2

T_1

T_3

T_2, T_3, T_4

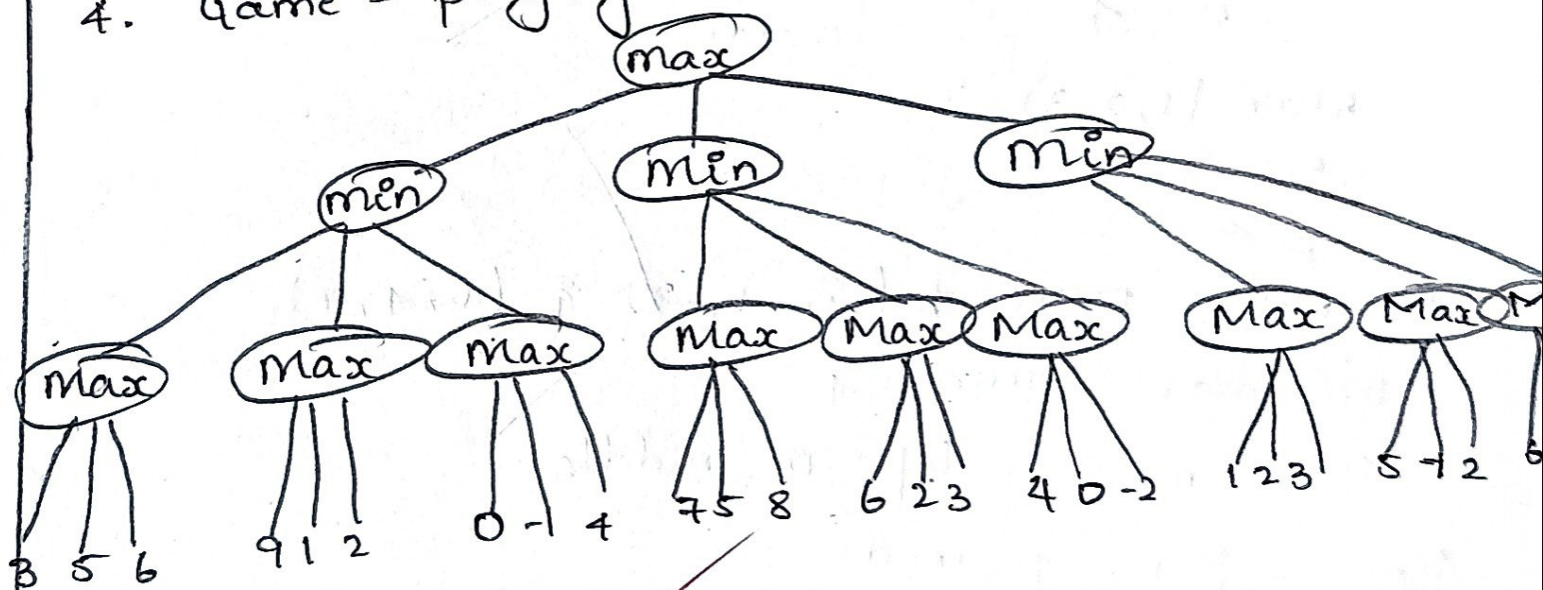
Step 6 $E_6 = T_2$

E_6 conflicts only with E_1 , which is at T_1

© final timetable

time slot	Examination	Hall	Students
T_1	E_1	A	100
T_1	E_4	C	60
T_2	E_2	B	80
T_2	E_3	A	70
T_2	E_6	C	50
T_3	E_5	C	50
T_4	--	B	90

4. Game - playing AI



Minimax

$$\min(6, 9, 4) = 4$$

$$\min(8, 6, 4) = 4$$

$$\min(3, 5, 7) = 3$$

Root max decision

$$\max(4, 4, 3) = 4$$

AI can choose either the left or middle move, both giving minimax value of 4

Alpha - Beta Pruning

$$\max(3, 5, 6) = 6$$

$$\beta = 6$$

$$\max(9, 1, 2)$$

$$\alpha = 9, \beta = 6$$

$$\alpha \geq \beta$$

remaining values 1 & 2 are pruned

$$\max(0, -1, 4) = 4$$

$$\min(6, 9, 4) = 4$$

$$\alpha = 4$$

$$\max(7, 5, 8) = 8$$

$$\max(6, 2, 3) = 6$$

$$\max(4, 0, -2) = 4$$

$$\min(8, 6, 4) = 4$$

$$\alpha = 4$$

$$\max(1, 2, 3) = 3$$

$$\beta = 3$$

$$\beta \leq \alpha$$

Branches pruned (5, -1, 2) & (6, 4, 7)

Minimax value = 4

Best move = left or middle

Alpha - Beta pruned

• value 1, 2 from second max node of left subtree

• Entire branches (5, -1, 2) & (6, 4, 7) from right subtree.

Delivery Route Optimization

Algorithm: Hill Climbing

Given neighbouring Routes:

$$S_1 = 26 \text{ km}$$

$$S_2 = 30 \text{ km}$$

$$S_3 = 24 \text{ km}$$

$$S_4 = 27 \text{ km}$$

$$S_5 = 23 \text{ km}$$

$$S_6 = 29 \text{ km}$$

Step 1:

Choose the best neighbour of initial ~~sol~~
solⁿ:-

$$\min(26, 30, 24, 27, 23, 29) = 23 \text{ km}$$

So choose ~~S_5~~

from S_5 , the best neighbouring solution has cost 22 km

All other neighbours have cost greater than 22 km

$\therefore$ Hill climbing stops because no better neighbour is available

final cost = 22 km

final solution is a local optimum.